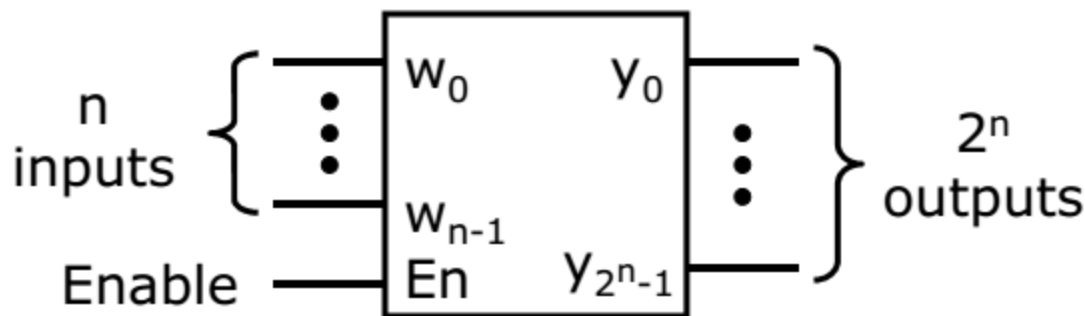


Combinatorial Circuit Building Blocks:

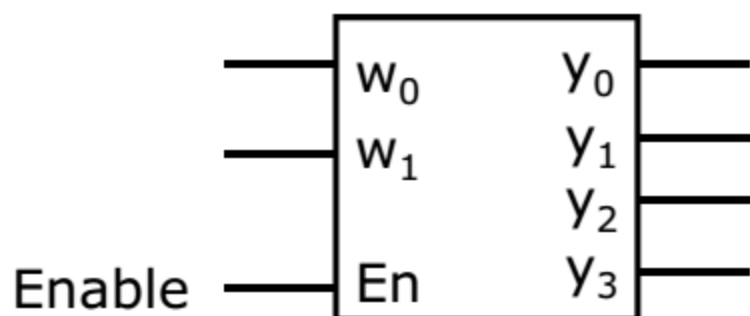
Decoders, Demultiplexers,
Encoders and Code Converters

Decoders

- Decoder circuits: decode encoded information
- A binary decoder has **n** data inputs and **2^n** outputs
- Only one output is asserted at any time (***one-hot encoded***) and each output corresponds to one valuation of the inputs
- An enable input (E_n) is used to disable the outputs
 - If $E_n=0$, none of the decoder outputs is asserted
 - If $E_n=1$, one of the outputs is asserted according to the valuation of the inputs

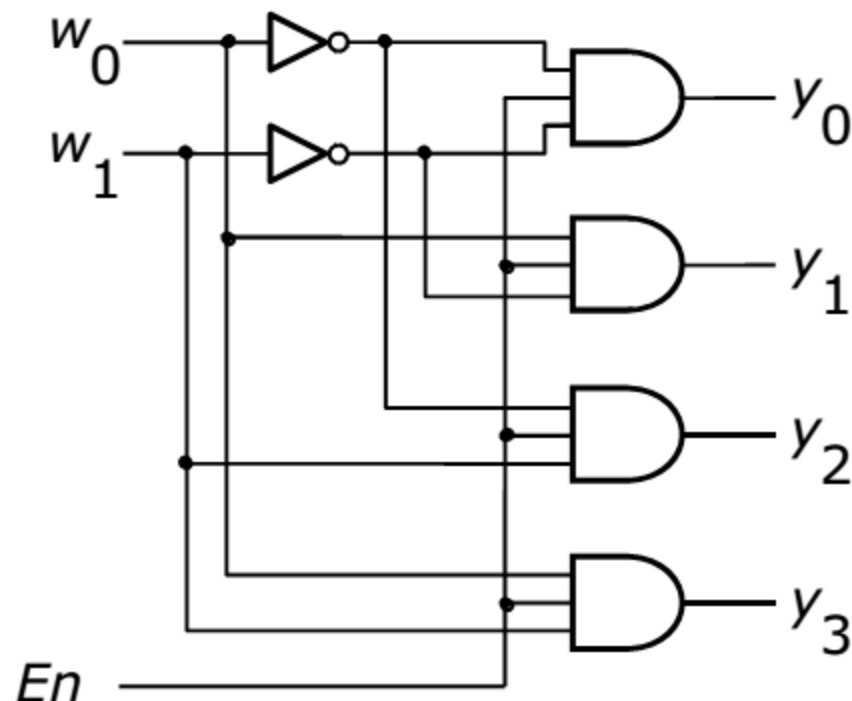


2-to-4 decoder circuit

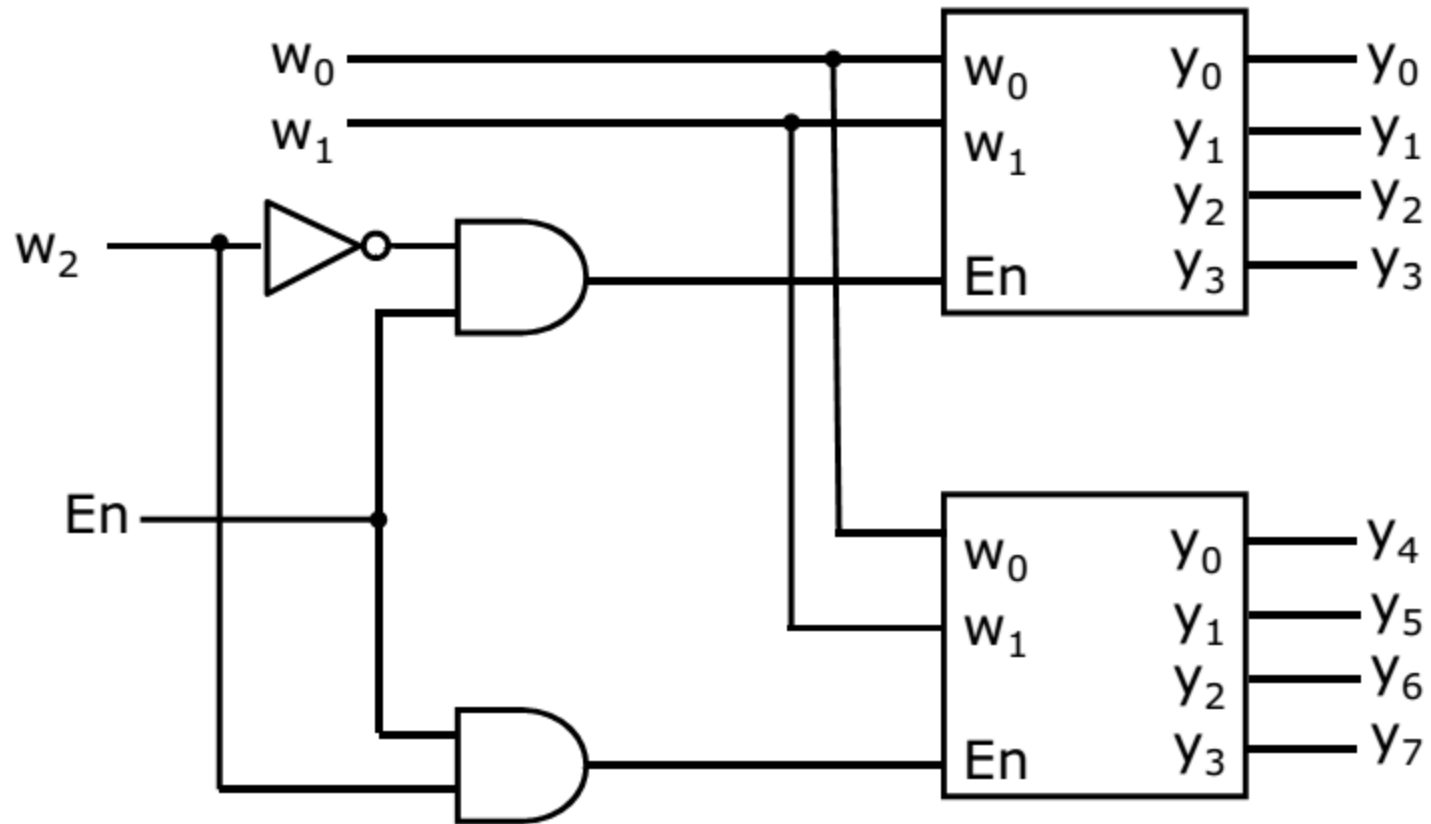


En	w_1	w_0	y_0	y_1	y_2	y_3
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1
0	X	X	0	0	0	0

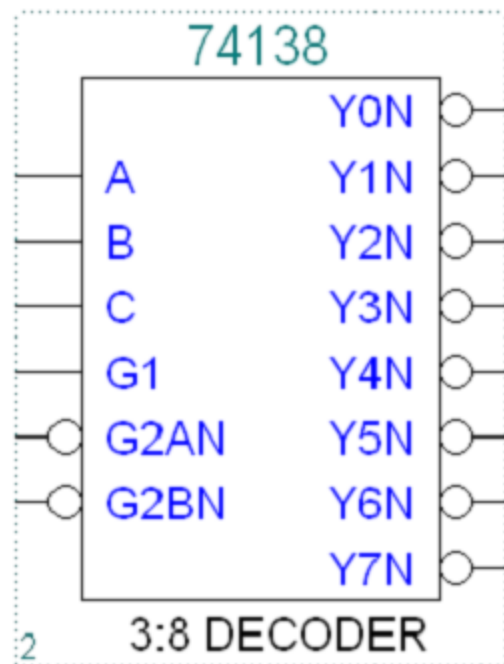
Truth table



3-to-8 decoder



74138 3-to-8 decoder



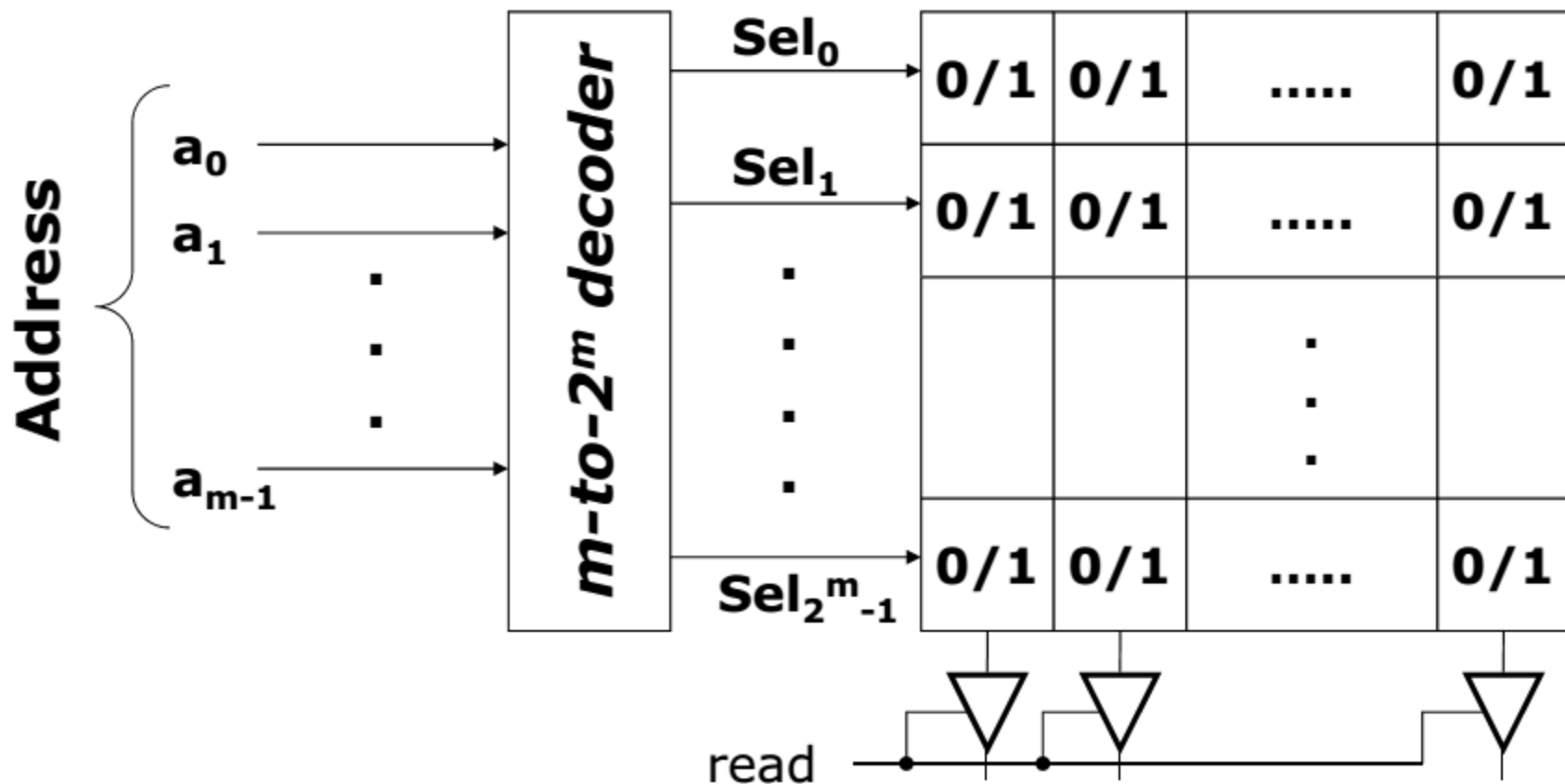
Inputs						Outputs							
Enable		Select											
G1	G2*	C	B	A		Y0N	Y1N	Y2N	Y3N	Y4N	Y5N	Y6N	Y7N
X	H	X	X	X		H	H	H	H	H	H	H	H
L	X	X	X	X		H	H	H	H	H	H	H	H
H	L	L	L	L		L	H	H	H	H	H	H	H
H	L	L	L	H		H	L	H	H	H	H	H	H
H	L	L	H	L		H	H	L	H	H	H	H	H
H	L	L	H	H		H	H	H	L	H	H	H	H
H	L	H	L	L		H	H	H	H	L	H	H	H
H	L	H	L	H		H	H	H	H	H	L	H	H
H	L	H	H	L		H	H	H	H	H	L	H	H
H	L	H	H	H		H	H	H	H	H	H	L	H

* $G2 = G2AN + G2BN$

Note the 'active low' outputs

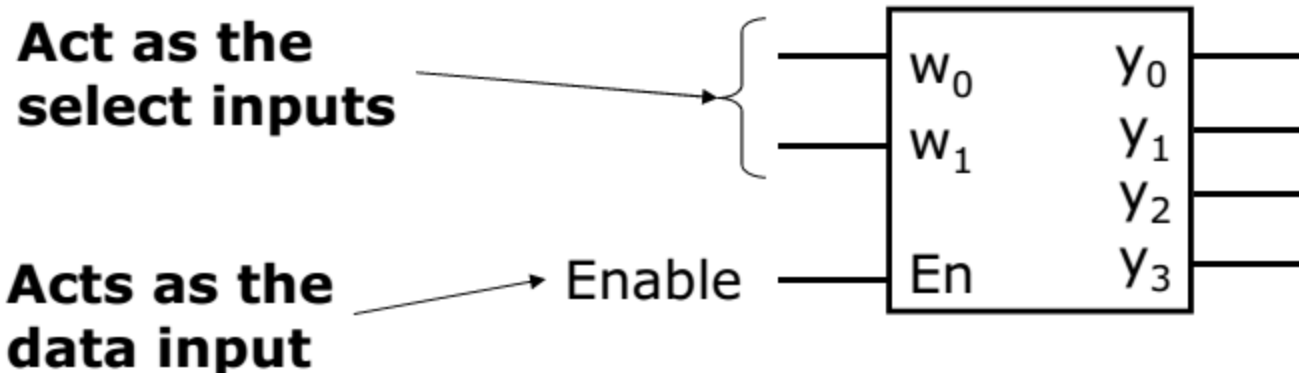
Decoder application

- A common decoder application is the decoding of address lines for memory chips



Demultiplexers

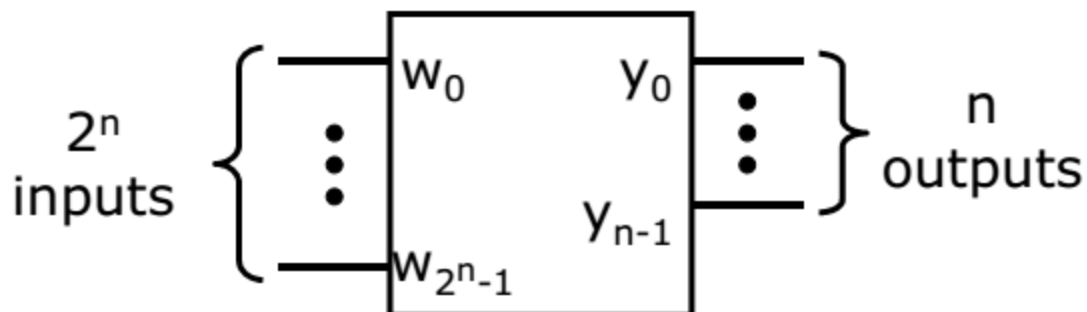
- A multiplexer multiplexed n data inputs to a single output
- A circuit that performs the opposite, placing the value of a single input onto one for multiple outputs is called a **demultiplexer**
- An n -to- 2^n decoder implements a 1-to- n demultiplexer



Encoders

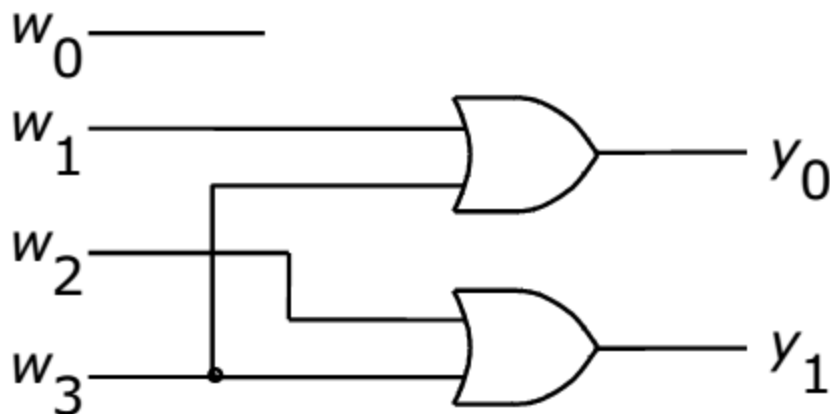
- An **encoder** performs the opposite function of a decoder
- A **binary encoder** encodes information (data) from 2^n inputs into an n -bit code (output)
 - Exactly one of the inputs should have a value of one
 - The outputs represent the binary number that identifies which input is equal to 1
- Encoders reduce the number of bits needed to represent given information
- Practical use: transmitting information in a digital system

Encoders



A 2^n -to- n binary encoder

w_3	w_2	w_1	w_0	y_1	y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1



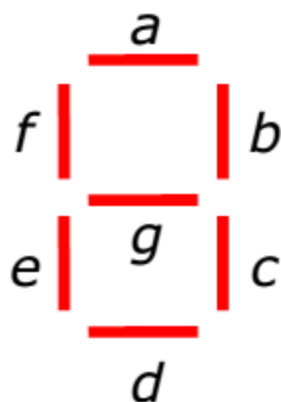
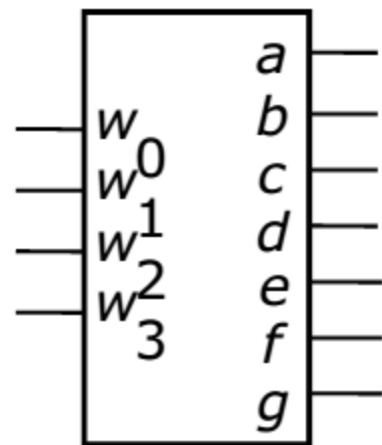
Code converters

- The purpose of **code converter** circuits is to convert from one type of input encoding to another type of output encoding
- For example:
 - A 3-to-8 decoder converts from a binary number to a one-hot encoding at the output
 - A 8-to-3 encoder performs the opposite
- Many different types of code converter circuits can be constructed
 - One common example is a BCD-to-7-segment decoder

BCD-to-7-segment decoder

- Converts one binary-coded decimal (BCD) digit into information suitable for driving a digit-oriented display
 - A vending machine display is an example
- The circuit converts a BCD digit into 7 signals that are used to drive (activate) the segments in the display
 - Each segment is a small light-emitting diode (LED), which glows when driven by an electrical signal

BCD-to-7-segment decoder



w_3	w_2	w_1	w_0	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1

BCD-to-7-segment decoder

